

Standalone Probit

probit family, scored against the standalone Softmax on the same utilities on 2015-2026, 5 seed(s), from unrecorded commit

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Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Standalone Probit probit	COMPARATOR the standalone Softmax on the same utilities	SEEDS 5 7, 17, 42, 123, 256	TEST YEARS 12 2015-2026
BOOSTING ROUNDS n/a	SOURCE COMMIT n/a clean		

Config digest unrecorded, model view bagged_probit, artifacts read from /Users/jlcheung/PycharmProjects/hkjc-mod/results/probit_models.

ENVIRONMENT

Component	Version
numpy	2.5.2
pandas	3.0.5
python	3.12.11
xgboost	3.4.1

Deltas in this report use model_minus_market_log_loss: race_multinomial_log_loss, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Executive summary

MEAN DELTA -0.012557 negative is better	YEARS BETTER 11/11	WORST YEAR -0.006424	CALIBRATED DELTA -0.007069 both sides calibrated
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Standalone Probit is better than its comparator by 0.012557 race_multinomial_log_loss, measured against the standalone Softmax on the same utilities on the evaluated races. The direction holds in every one of the 11 evaluated years. It remains behind the market in absolute terms. Beating its comparator and beating the market are different claims and only the first is made here. Every seed agrees on the sign of the improvement.

Comparator and evaluation population

Softmax, not the market and not the Offset model. The question this answers is narrow and has to stay narrow: does replacing independent Gumbel noise with correlated normal noise help, holding the mean utilities fixed? Scoring it

against the Offset model would compare two different mean learners and one link change at the same time.

TEST RACES

8,675

YEARS

12

2015-2026

SEEDS

5

HONEST-SCALE YEARS

11

years whose utility scale was fitted rather than fallen back

Some years fell back rather than fitting their own scale

2015 did not produce an honest utility scale (insufficient_history). Those years are excluded from the headline delta and from the bankroll paths, and shown separately in the year-by-year table: a fallback scale is not a measurement of the candidate.

Overall probability performance

races	model log loss	comparator log loss	market log loss	honest delta	better year share
43,375	2.030823	2.043380	2.028805	-0.012557	1

One row, pooled over every evaluated race. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	comparator log loss	market log loss	delta	honest
2015	3710	2.220620	2.020336	1.998720	0.200284	no
2016	3720	2.098827	2.105251	2.099277	-0.006424	yes
2017	3860	2.023286	2.036819	2.021559	-0.013534	yes
2018	3780	2.073354	2.081233	2.061504	-0.007879	yes
2019	3795	1.935353	1.952049	1.932827	-0.016696	yes
2020	3995	2.012171	2.030497	2.010400	-0.018326	yes
2021	3915	2.011146	2.026374	2.006659	-0.015228	yes
2022	3915	2.059349	2.066103	2.051195	-0.006754	yes
2023	3980	2.022700	2.032808	2.024066	-0.010108	yes
2024	3960	2.020289	2.034578	2.025744	-0.014289	yes
2025	3740	2.059159	2.067962	2.057397	-0.008802	yes
2026	1005	2.023421	2.043510	2.026224	-0.020089	yes



Ahead of its comparator in 11 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: candidate and comparator are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. Fallback-scale years are excluded here, as they are from the headline.

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2016	2.096131	2.103251	-0.007120
2017	2.024009	2.031793	-0.007785
2018	2.073354	2.079390	-0.006036
2019	1.938096	1.945433	-0.007337
2020	2.012574	2.024462	-0.011888
2021	2.011085	2.017480	-0.006395
2022	2.059959	2.065483	-0.005524
2023	2.023223	2.025731	-0.002508
2024	2.021114	2.026537	-0.005423
2025	2.060147	2.066830	-0.006683

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2026	2.024664	2.035721	-0.011056

If the calibrated delta is materially smaller than the uncalibrated one, part of the candidate's advantage was a temperature the comparator could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement compares a model's first pick against the market's. This candidate is scored against its own comparator on fixed mean utilities, not against the market, so a market top-choice comparison would answer a different question than the rest of this report.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is a choice **link** applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects to attribute. The link changes how utilities become probabilities; the utilities themselves come from the comparator's fit, whose feature effects are reported in that model's own report.

Representative races

Not applicable to this model. Representative races are chosen by how much a model's correction moved a runner's probability. This candidate applies a link to fixed mean utilities, so the movement to explain belongs to the comparator's fit and is shown in that model's report. A race chosen here would be chosen by integration noise.

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
flat_stake@edge>=0.02	6510	65,100,000	-25,665,000	-0.394240	-36.841361	2.531206	-15,665,000	3833
kelly_no_rebate@0.01	7363	15,537,960	-2145786.500000	-0.138100	-0.241535	0.221390	7854213.500000	4256
kelly_settled_rebate@0.01	7369	16,228,700	-2,254,076	-0.138894	-0.162460	0.161965	8,500,497	4254
rebate_aware_kelly@0.01	7715	18,207,680	-2,607,460	-0.143207	-0.191536	0.188440	8256902.450000	4251

Model view bagged_probit. Seed -1. Every rule is settled on the same selected bets at the same odds, so the differences between these rows are stake sizing alone.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE (EDGE>=0.02)

bucket lower	bucket upper	n bets	mean estimated edge	expected return	realized return
0.020000	0.050000	951	0.034377	0.034377	-0.173817
0.050000	0.100000	1002	0.072830	0.072830	0.014271
0.100000	0.200000	1230	0.144020	0.144020	-0.353780
0.200000	unbounded	3327	0.578110	0.578110	-0.595236

KELLY, NO REBATE (0.01)

bucket lower	bucket upper	n bets	mean estimated edge	expected return	realized return
0	0.020000	853	0.010508	0.010508	-0.213567
0.020000	0.050000	951	0.034377	0.034377	-0.004506
0.050000	0.100000	1002	0.072830	0.072830	-0.209183
0.100000	0.200000	1230	0.144020	0.144020	-0.096589
0.200000	unbounded	3327	0.578110	0.578110	-0.146942

KELLY, SETTLED REBATE (0.01)

bucket lower	bucket upper	n bets	mean estimated edge	expected return	realized return
0	0.020000	859	0.010483	0.010483	-0.209815
0.020000	0.050000	951	0.034377	0.034377	0.000765
0.050000	0.100000	1002	0.072830	0.072830	-0.209753
0.100000	0.200000	1230	0.144020	0.144020	-0.100818
0.200000	unbounded	3327	0.578110	0.578110	-0.147501

REBATE-AWARE KELLY (0.01)

bucket lower	bucket upper	n bets	mean estimated edge	expected return	realized return
unbounded	0	351	-0.031497	-0.031497	-0.310676
0	0.020000	854	0.010503	0.010503	-0.228985
0.020000	0.050000	951	0.034377	0.034377	0.016299
0.050000	0.100000	1002	0.072830	0.072830	-0.211816
0.100000	0.200000	1230	0.144020	0.144020	-0.085301
0.200000	unbounded	3327	0.578110	0.578110	-0.148832

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts here are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

KELLY PRESETS

OPENING BANKROLL

HK\$10,000,000

PRESETS SETTLED

1% Kelly

PATH MODES

annual reset,
continuous

SEEDS

5

7, 17, 42, 123, 256

YEARS STAKED

2015-2026

Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. The bankroll config declares 2% Kelly, 5% Kelly, 10% Kelly as well, and this run did not settle them, so they have no button here: a preset that showed nothing would be worse than one that is absent. 2015 is excluded from every path below. These test years have no earlier year to fit the utility scale on and fall back to the predeclared theoretical value. The utility is divided by the scale, so the fallback amplifies corrections where a fitted scale halves them: their stakes and claimed edges are several times any fitted year's and are not comparable with them. Exclude these years from any return figure -- filter on is_honest_scale in the per-run CSV.

1% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

-0.116%

Kelly, no rebate, annual reset

BETS

6,121

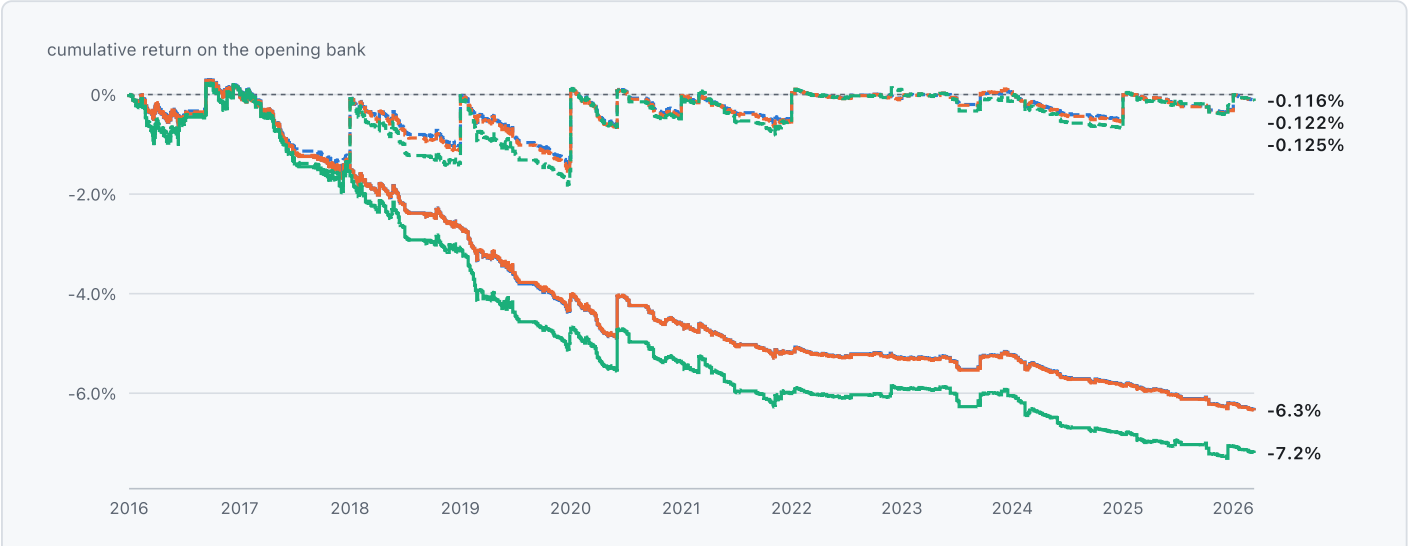
settled bets

REBATE RECEIVED

HK\$62,394

the most any rule received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	3,525	-0.116%	6,121 bets
Kelly, no rebate, continuous	3,525	-6.3%	6,121 bets
Kelly, settled rebate, annual reset	3,528	-0.125%	6,127 bets
Kelly, settled rebate, continuous	3,528	-6.3%	6,127 bets
Rebate-aware Kelly, annual reset	3,525	-0.122%	6,283 bets
Rebate-aware Kelly, continuous	3,525	-7.2%	6,283 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	751	751	773
2017	881	883	911
2018	789	789	811
2019	1,038	1,038	1,068
2020	964	965	980
2021	570	571	583
2022	279	279	285
2023	230	230	236
2024	298	299	308

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2025	261	261	267
2026	60	61	61

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 751 bets in 2016 against 261 in 2025. Kelly, settled rebate: 751 bets in 2016 against 261 in 2025. Rebate-aware Kelly: 773 bets in 2016 against 267 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$62,394; Rebate-aware Kelly, continuous HK\$62,394; Kelly, settled rebate, annual reset HK\$4,864; Kelly, settled rebate, continuous HK\$4,864. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,988,420	-0.116%	6,121	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$8,385,931	HK\$7,854,214	-6.3%	6,121	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,987,516	-0.125%	6,127	HK\$4,864
1% Kelly	Kelly, settled rebate, continuous	HK\$9,076,780	HK\$8,500,497	-6.3%	6,127	HK\$4,864
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,987,824	-0.122%	6,283	HK\$62,394
1% Kelly	Rebate-aware Kelly, continuous	HK\$8,897,664	HK\$8,256,902	-7.2%	6,283	HK\$62,394

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

MEAN DELTA
-0.012063

INTERVAL
-0.0140 to
-0.0101
80% level

EXCLUDES ZERO
FAVOURABLY
yes

RACE DAYS
891
39,665 races

REPLICATES
2,000
seed 20260824

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix

THE UTILITY SCALE

FITTED MINIMUM	FITTED MAXIMUM	THEORY
1.3893	1.4534	0.7797
		the value theory predicts

The fitted scale sits above the value theory predicts, in every year: 1.389 to 1.453 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit
Comparator field	softmax
Integration	exact_gauss_hermite_quadrature
Shared factors	0
Opening bankroll	10,000,000
Bankroll mode	continuous
Rebate-aware sizing	no

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	pass
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

6 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.013234
17	-0.012556
42	-0.012215
123	-0.012210
256	-0.012571

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

INTEGRATION SENSITIVITY

architecture	study	setting	value	metric	n races
probit	utility_scale	2015	0.779697	n/a	0
probit	utility_scale	2016	1.389269	n/a	0
probit	utility_scale	2017	1.453396	n/a	0
probit	utility_scale	2018	1.439805	n/a	0
probit	utility_scale	2019	1.450206	n/a	0
probit	utility_scale	2020	1.428605	n/a	0
probit	utility_scale	2021	1.426172	n/a	0
probit	utility_scale	2022	1.420903	n/a	0
probit	utility_scale	2023	1.428470	n/a	0
probit	utility_scale	2024	1.426704	n/a	0
probit	utility_scale	2025	1.423754	n/a	0
probit	utility_scale	2026	1.426958	n/a	0

12 rows from the sensitivity artifact. If the probabilities move materially with the integration settings, the result is a property of the integration rather than of the model.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Standalone Probit against the standalone Softmax on the same utilities. Beating a comparator and beating the market are different claims.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.